ALA-2:- Python Programming

Topic: Input and Output Functions

Objective

The objective of this activity is to understand and implement Python input and output functions through detailed interview-based questions. This helps in strengthening problem-solving skills and prepares students for real-world programming and technical interviews.

Introduction

Python provides simple and powerful functions to interact with users. The input() function is used to take input from the user, while the print() function is used to display output. These functions are fundamental and widely used in almost every Python program.

Q1. Write a Python program to take the user's name as input and display a greeting message.

```
name = input("Enter your name: ")
print("Hello", name)
```

Q2. Write a Python program to take two numbers from the user and display their sum.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Sum =", a + b)
```

Q3. Write a Python program to check whether a given number is even or odd.

```
num = int(input("Enter a number: "))
if num % 2 == 0:     print("Even")
else:     print("Odd")
```

Q4. Write a Python program to take input from the user and display its data type.

```
data = input("Enter something: ")
print("Data type is:", type(data))
```

Q5. Write a Python program to convert temperature from Celsius to Fahrenheit.

```
c = float(input("Enter temperature in Celsius: "))
f = (c * 9/5) + 32 print("Fahrenheit =", f)
```

Q6. Write a Python program to calculate simple interest based on user input.

```
p = float(input("Enter principal: "))
r = float(input("Enter rate: ")) t =
float(input("Enter time: ")) si = (p
* r * t) / 100 print("Simple Interest
=", si)
```

Q7. Write a Python program to swap two numbers entered by the user.

```
a = int(input("Enter a: "))
b = int(input("Enter b: "))
a, b = b, a
print("After swapping:", a, b)
```

Q8. Write a Python program to take multiple inputs in a single line and display their sum.

```
a, b = map(int, input("Enter two numbers: ").split())
print("Sum =", a + b)
```

Conclusion

This activity provided a clear understanding of Python input and output functions. By solving interview-based questions, students improved their logical thinking and practical coding skills.